CPE 381 Classwork 07: Building Blocks for Continuous-Time Signals

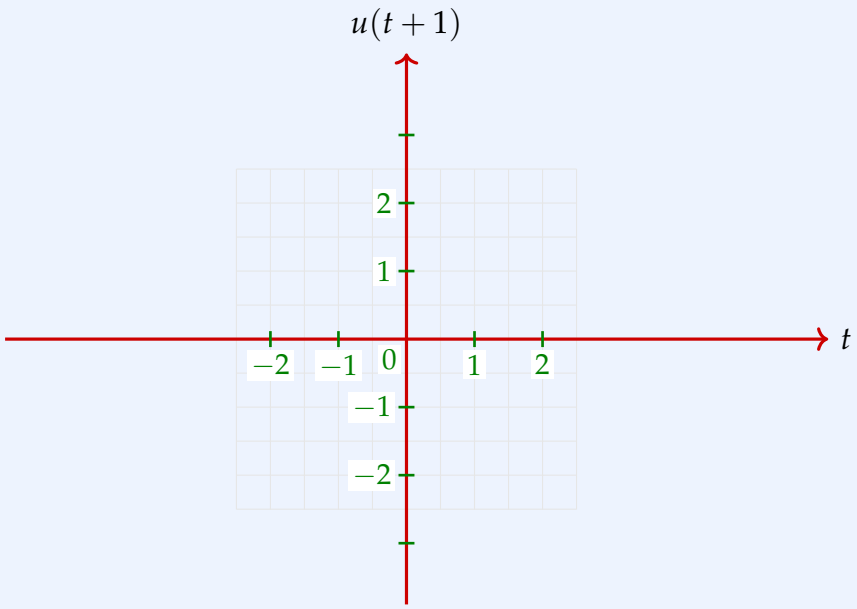
Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

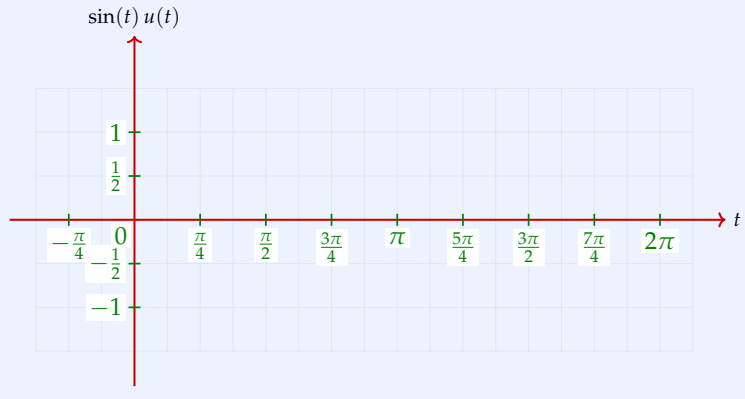
1. Assuming $u(t)$ is the unit-step function as discussed in the class, sketch $u(t+1)$.

Answer

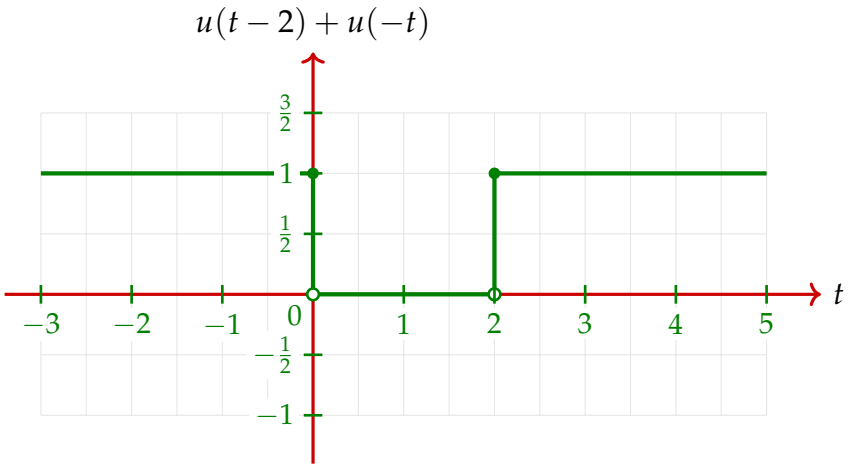


2. Sketch $x(t) = \sin(t)u(t)$.

Answer



3. Write mathematical equation corresponding to the following graph.



Answer

4. Calculate the energy of the signal

$$z(t) = \begin{cases} 1, & 0 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

Answer